Exercise Sheet 2

The Lie algebras are complex and finite-dimensional. Use the notation E_{ij}, N_r, $\mathfrak{n}_n$ and $C^k\mathfrak{g}$ from Lecture 2.

Exercise 2.1 (The algebra $\mathfrak{n}_n$). Let $n \geq 2$.

(i) Prove that $C^k\mathfrak{n}_n = N_{k+1}$ for $0 \leq k \leq n-1$. Deduce that $C^{n-2}\mathfrak{n}_n = \mathbb{C}E_{1n}$ and $C^{n-1}\mathfrak{n}_n = 0$.

(ii) Prove that $Z(\mathfrak{n}_n) = \mathbb{C}E_{1n}$.

Exercise 2.2 (Ideals meet the centre). Let $\mathfrak{g}$ be nilpotent and let $I \neq 0$ be an ideal of $\mathfrak{g}$.

(i) Prove that $I \cap Z(\mathfrak{g}) \neq 0$. Apply Engel's linear theorem to the restrictions of the operators $\operatorname{ad} x$ to I.

(ii) Deduce that every one-dimensional ideal of $\mathfrak{g}$ is central.

(iii) Show that the conclusion of (i) fails for the upper triangular algebra $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$ and $I = \mathbb{C}e$. Which hypothesis fails?

Exercise 2.3 (Extensions and flags of ideals).

(i) Show that $\mathfrak{b}$ has an ideal $\mathfrak{a}$ such that $\mathfrak{a}$ and $\mathfrak{b}/\mathfrak{a}$ are nilpotent, while $\mathfrak{b}$ is not. Why does this not contradict the lecture's lemma on central quotients?

(ii) Let $\mathfrak{g} \neq 0$ be nilpotent. Prove that $[\mathfrak{g},\mathfrak{g}] \neq \mathfrak{g}$, and that $\mathfrak{g}$ has an ideal of codimension one.

(iii) Let $n = \dim \mathfrak{g}$. Prove that $\mathfrak{g}$ is nilpotent if and only if there are ideals

$$0 = \mathfrak{g}_0 \subset \mathfrak{g}_1 \subset \cdots \subset \mathfrak{g}_n = \mathfrak{g}, \qquad \dim \mathfrak{g}_i = i, \qquad [\mathfrak{g},\mathfrak{g}_i] \subseteq \mathfrak{g}_{i-1}\ (1 \leq i \leq n).$$

Exercise 2.4 (The adjoint of a nilpotent matrix). Let $e = E_{12} \in \mathfrak{gl}_2(\mathbb{C})$. Use the ordered basis $(E_{11}, E_{12}, E_{21}, E_{22})$ of $\mathfrak{gl}_2(\mathbb{C})$.

(i) Compute the matrix D of $\operatorname{ad} e$ and its powers. Show that $(\operatorname{ad} e)^2 \neq 0$ and $(\operatorname{ad} e)^3 = 0$. Compare with the bound $2m-1$ of the lecture.

(ii) Prove that $\operatorname{ad}(x + cI) = \operatorname{ad} x$ for $x \in \mathfrak{gl}(V)$ and $c \in \mathbb{C}$. Give a non-nilpotent $x \in \mathfrak{gl}_2(\mathbb{C})$ with $\operatorname{ad} x$ nilpotent.

Exercise 2.5 (Limits of the matrix criterion). Use the basis (h, e, f) of $\mathfrak{sl}_2$ from Lecture 1.

(i) Show that e and f are nilpotent but have no common null vector in $\mathbb{C}^2$. Which hypothesis of Engel's linear theorem fails for $\operatorname{span}(e, f)$?

(ii) Let $u = h + e - f$. Show that (e, f, u) is a basis of $\mathfrak{sl}_2$ consisting of ad-nilpotent elements. Why does this not contradict the abstract form of Engel's theorem?

(iii) Let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation with $V \neq 0$, every $\rho(x)$ nilpotent, and $\ker \rho \subseteq Z(\mathfrak{g})$. Prove that $\mathfrak{g}$ is nilpotent. Show that the condition $\ker \rho \subseteq Z(\mathfrak{g})$ cannot be omitted.

Exercise 2.6 (Nilpotent algebras of dimension at most three).

(i) Prove that every nilpotent Lie algebra of dimension at most two is abelian.

(ii) Prove that every nonabelian nilpotent Lie algebra of dimension three is isomorphic to $\mathfrak{h}_3$.